IC101

Introductory Securities & Markets

Lecture 3 · Commercial, Investment and Central Banks

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From last week to this week

We found the fragile balance sheet. Now: how it works, why it breaks.

Last week: we X-rayed a bank

A company like Apple runs on ~20% equity. A retailer 25 to 40%.

But a bank runs on only ~5% equity. It is almost entirely OTHER PEOPLE'S money.

We left it on a cliffhanger: that thin cushion is why banks are special, and why they collapse.

This week: three questions

1. What does a bank actually DO with your money?
2. Why does it BREAK, and what is a bank run?
3. Who STOPS it breaking?

Then the three types of bank fall out of the answers.

Bottom line: SVB, 2023, and Northern Rock, 2007, both died the same way. Today is about exploring how.

Today: the life and death of a bank

One story, not six separate topics: each step leads into the next.

What a bank really is — Licensed to create deposits, and those deposits are the nation's money.

The magic trick — What it DOES with them: turns short, safe savings into long, risky loans.

Why it is fragile — And why that trick is dangerous: fixed promises on assets that can fall.



The bank run — When the danger strikes: how a sound bank can die in a single day.

The safety net — What stops it: capital, deposit insurance, and the central bank behind it all.

The other bank — A different animal: investment banks live in the markets, not on deposits.

What is a bank, really?

Not a vault. An institution licensed to take deposits, and so to create money.

The legal definition

A bank is an institution with a licence to take DEPOSITS.

Your deposit is your asset, but the bank's liability, it owes you the money.

And here is the strange part: those deposits ARE the country's money. About 97% of all money is just bank deposits.

Why banks are 'special'

Fixed value: £1 in the bank is always worth £1.

On demand: you can take it out any time.

They run the payments system: cards, transfers, salaries.

They provide the credit that lubricates the whole economy.

No wonder they are licensed and watched so closely.

~97%
of all money is bank deposits

~$190tn
global banking assets

~$6tn
ICBC, the world's biggest bank

4 of 5
biggest banks are Chinese

Money split: Bank of England. Bank size: S&P Global / The Banker, 2024-25.

The magic trick: transformation

A bank turns short, safe, small savings into long, risky, large loans



Maturity transformation borrow short, lend long. It profits from the gap, but must always find cash for today's withdrawals.

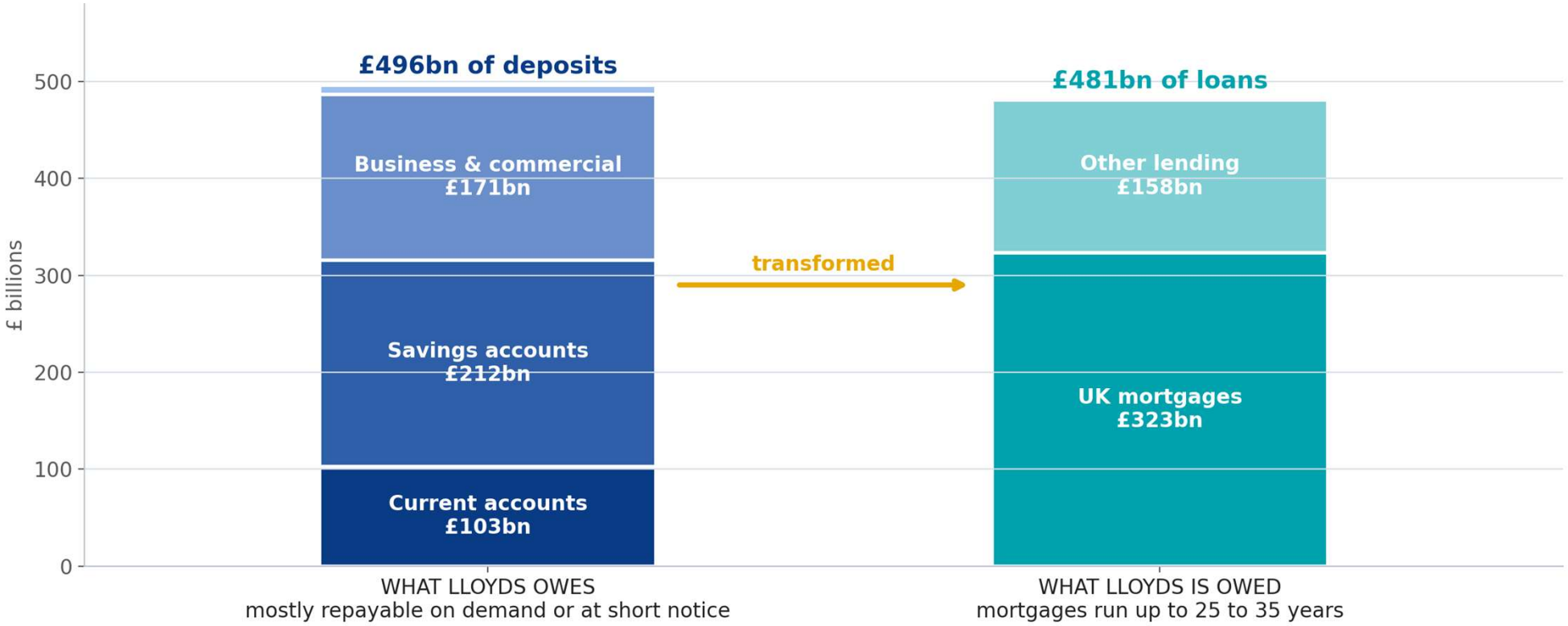
Risk transformation each loan is risky, but spread across thousands (law of large numbers) the pool is predictable.

Bottom line: this transformation is what a bank IS, and exactly why it is fragile: it lives on a promise it cannot instantly keep.

The goldsmiths of 17th-century London invented this: lending out deposited gold, keeping only a fraction in reserve.

The magic trick, in Lloyds' real numbers

£496bn that can leave at short notice is funding £481bn that comes back over decades



Source: Lloyds Banking Group 2025 results, 31 December 2025. Savings includes some fixed-term deposits.

Takeaway: This is maturity transformation in one real bank. Short, safe deposits on the left fund long, risky loans on the right. The gap between the two is where the profit comes from, and where the danger lives.

Quick poll, grab your phones

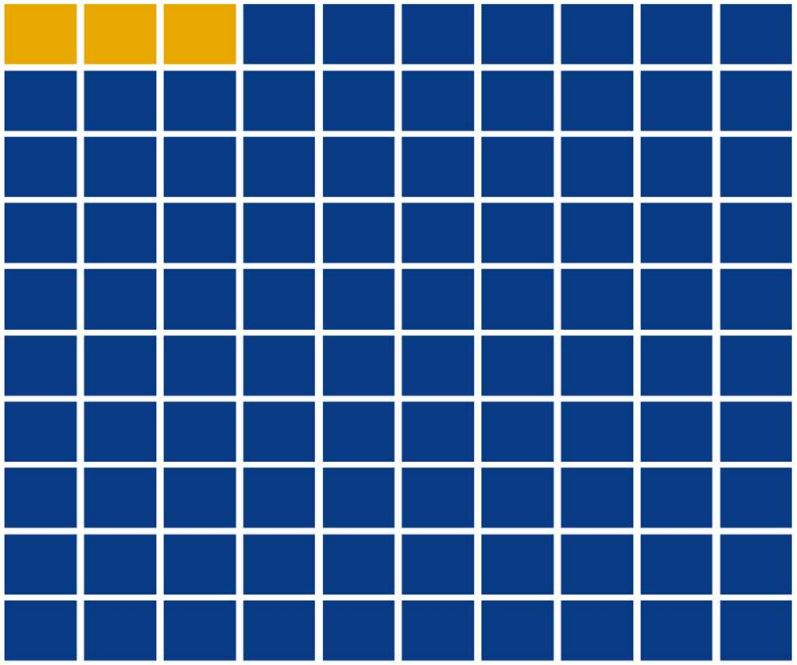
You deposit £1,000 in your bank. Where is that money right now?

- **A** Sitting in the bank's vault, waiting for you
- **B** Lent out, as someone's mortgage or business loan
- **C** Held at the Bank of England
- **D** Turned into cash in an ATM

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Where the UK's money actually is

97 of every 100 pounds exist only as entries in a bank's ledger



Of every £100 of money in the UK:

£3 is notes and coin, issued by the Bank of England

£97 is bank deposits, numbers in banks' ledgers

Total money (M4): about £3.3 trillion

Notes in circulation: about £97 billion

Where do the deposits come from? Mostly from commercial banks lending: a bank creates a deposit every time it makes a loan. Some came from the Bank of England's QE, which paid for gilts by crediting investors' accounts.

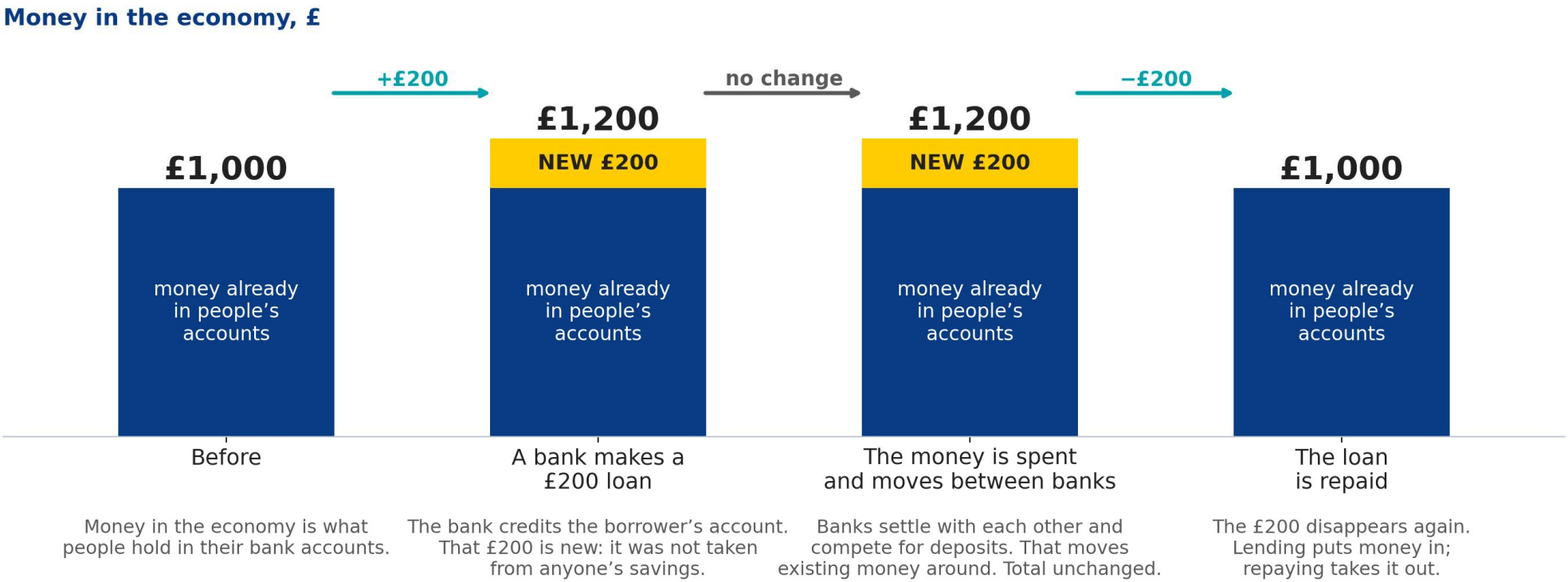
Sources: Bank of England, M4 August 2026 (£3,306bn); notes in circulation (weekly report, March 2026); 'Money creation in the modern economy' (2014). Shares rounded.

Takeaway: Only about £97bn of the £3.3tn is physical cash. Most of the rest was created by commercial banks making loans, and some by the Bank of England's QE. Either way, banks' ledgers, not the mint, are where money lives.

CLICK How do banks create money? Ask AI ↗

How bank lending creates money

A loan adds money. Moving it between banks does not. Repaying removes it.



Banks do not print money. They create it by lending, and they cancel it when loans are repaid.

The money in the economy is the running total of lending minus repayments. **It is not unlimited:** needs creditworthy borrowers, a loan has to be profitable, capital rules cap it (10% equity) and the cash to settle it costs base rate.

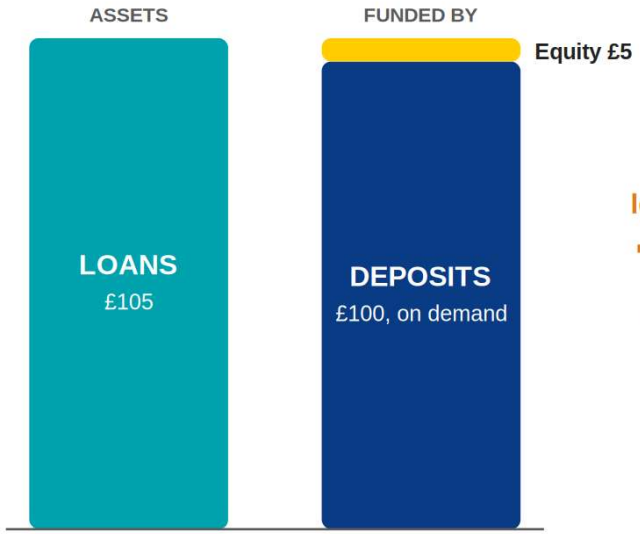
That is why the cost of borrowing is how the central bank steers the money supply (**low rates → more borrowing → increase in money supply**)

Why banks are fragile

Fixed promises on one side, assets that can fall on the other

A normal day

the loans cover the deposits, with £5 to spare



loans lose 6%

a bad year: some borrowers default

The same bank, one bad year later

the deposits are still £100, but the loans no longer cover them



insolvent: it owes more than it owns

Three fragilities

1. Leverage

Only £5 of equity under £100 of deposits. A 5% fall in the loans wipes the owners out.

2. Maturity mismatch

Deposits are due on demand. The loans are locked up for years.

3. Falling assets

The £100 owed is fixed. The loans can lose value whenever borrowers default.

Fixed liabilities, variable assets, thin cushion.

Pricing the risk: expected loss

*Banks can't know **WHICH** loans go bad for sure, but they know how many will*

Expected loss = PD × LGD × amount

PD probability of default: the chance this borrower fails

LGD loss given default: how much you lose if they do

Worked example

A £1,000 loan. History says this type of borrower defaults 2% of the time (PD) and the bank loses 40% when they do (LGD).

Expected loss = 2% × 40% × £1,000 = £8.

The key idea

The bank builds that £8 into the interest rate, so EXPECTED losses are paid out of revenue.

It is the UNEXPECTED losses, the bad year worse than the average, that eat into equity.

CLICK **Explain expected loss (PD x LGD) with AI** ↗

Scott-Quinn, Ch 8. 'Expected losses are met out of revenue; unexpected losses are taken by shareholder funds.'

Quick poll, grab your phones

A bank has £100 of deposits and £5 of equity. Its loans fall 6% in value. What happens?

- **A** Nothing, the bank easily absorbs it
- **B** The owners' equity is wiped out and the bank is insolvent
- **C** Depositors instantly lose all their money
- **D** The government automatically pays

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Who absorbs the losses? Equity first

The same waterfall as Lecture 2, but now the cushion is tiny

Losses eat from the left. How far they get decides who pays.

A 2% loss: absorbed by this year's profit. Nobody notices.

A 6% loss: revenue gone, equity down to £1. Owners badly hurt, depositors safe.

A 10% loss: equity wiped out, depositors lose £3. Insolvent.



not to scale

Layer 1. Expected losses are priced into the interest rate, so revenue pays them.

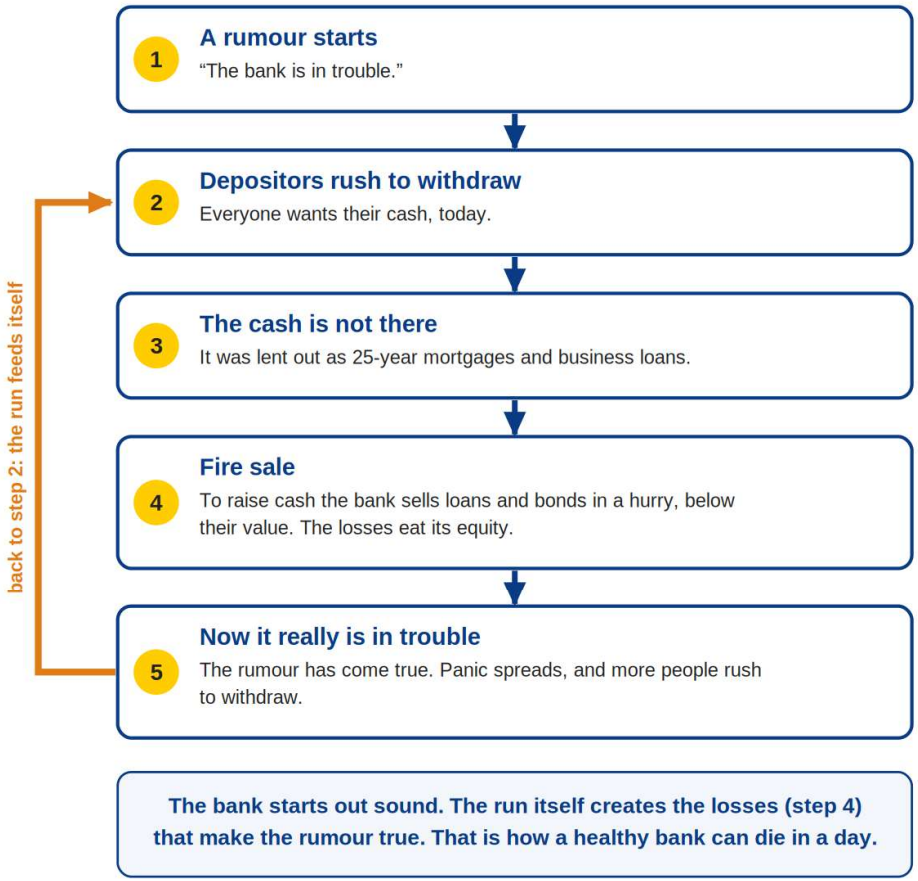
Layer 2. Unexpected losses, the bad year, come out of equity. Only £5 of it.

Layer 3. Once equity is gone, the 'safe' money takes the hit. The bank has failed.

Bottom line: **Expected losses come out of revenue. Unexpected losses come out of equity. With only ~5% equity, it does not take much.**

The bank run: how a sound bank dies in a day

Everyone wants their money at once, and it simply isn't there



The bank starts out sound. The run itself creates the losses (step 4) that make the rumour true. That is how a healthy bank can die in a day.

It really happened: Northern Rock, September 2007

£2bn
Withdrawn in three days, after the BBC reported that it had asked the Bank of England for emergency help.

~140 years
Since the last run on a British bank (Overend, Gurney, 1866).

Feb 2008
Nationalised. The queues down the high street became the defining image of the crisis.

The mechanism is 300 years old. The speed is new.

Northern Rock's run took days: queues, branch by branch. SVB's run in 2023 took hours: withdrawals by app, panic by social media. Same five steps, now at the speed of a tap. (Next slide.)

The new bank run: at the speed of a tap

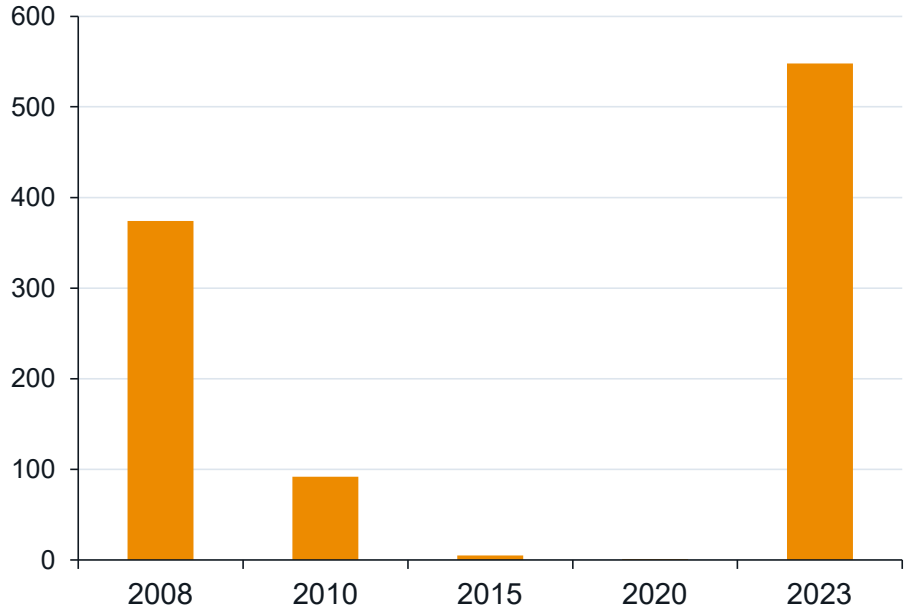
Social media and mobile banking made 2023 unlike anything before

Then vs now

2007, Northern Rock: DAYS. Physical queues, branch by branch.

2023, Silicon Valley Bank: HOURS. $42bn gone in a single day, another ~$100bn queued for day two. About 85% of deposits, on phones, sparked on group chats and X.

Assets of failed US banks $ billions



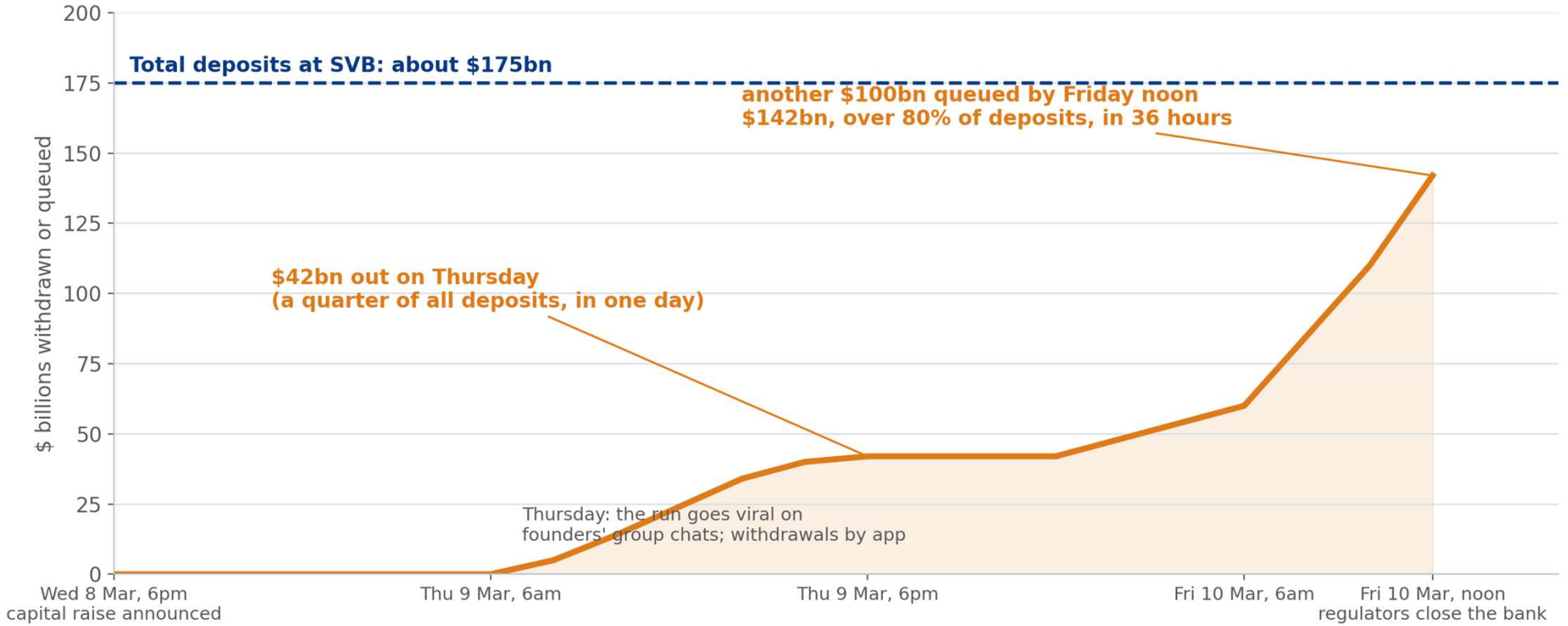
Bottom line: 2023 saw more bank assets fail than in 2008. Deposits now flee faster than the rulebook ever assumed, so regulators are rewriting it for the smartphone age.

DEEPER READ Silicon Valley Bank collapse: how it happened (CNBC) ↗

US failed-bank assets: FDIC. SVB run figures: Federal Reserve (Barr testimony), 2023.

SVB: $142bn tried to leave in 36 hours

The 2023 run, hour by hour



Sources: Federal Reserve (Vice Chair Barr testimony, March 2023); FDIC. Curve between the reported points is illustrative.

Takeaway: Northern Rock's queues took days. SVB's run happened on phones, in hours: $42bn gone on the Thursday and another $100bn queued by Friday noon. Nothing in the rulebook assumed money could move this fast.

Quick poll, grab your phones

Two banks are failing. Which should the central bank rescue?

- **A** Bank A: sound loans, but hit by a panic run (illiquid)
- **B** Bank B: made terrible loans, now worth less than it owes (insolvent)
- **C** Both of them
- **D** Neither, let the market decide



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Illiquidity vs insolvency

The distinction that decides whether a bank is saved or left to fail

Illiquid

Sound, but cannot find cash TODAY.

Its assets are good, they are just locked up long while depositors want cash now.

A cash-flow problem. A loan could save it.

Insolvent

Genuinely bust: owes more than it owns.

Its loans have gone bad; assets are worth less than its deposits.

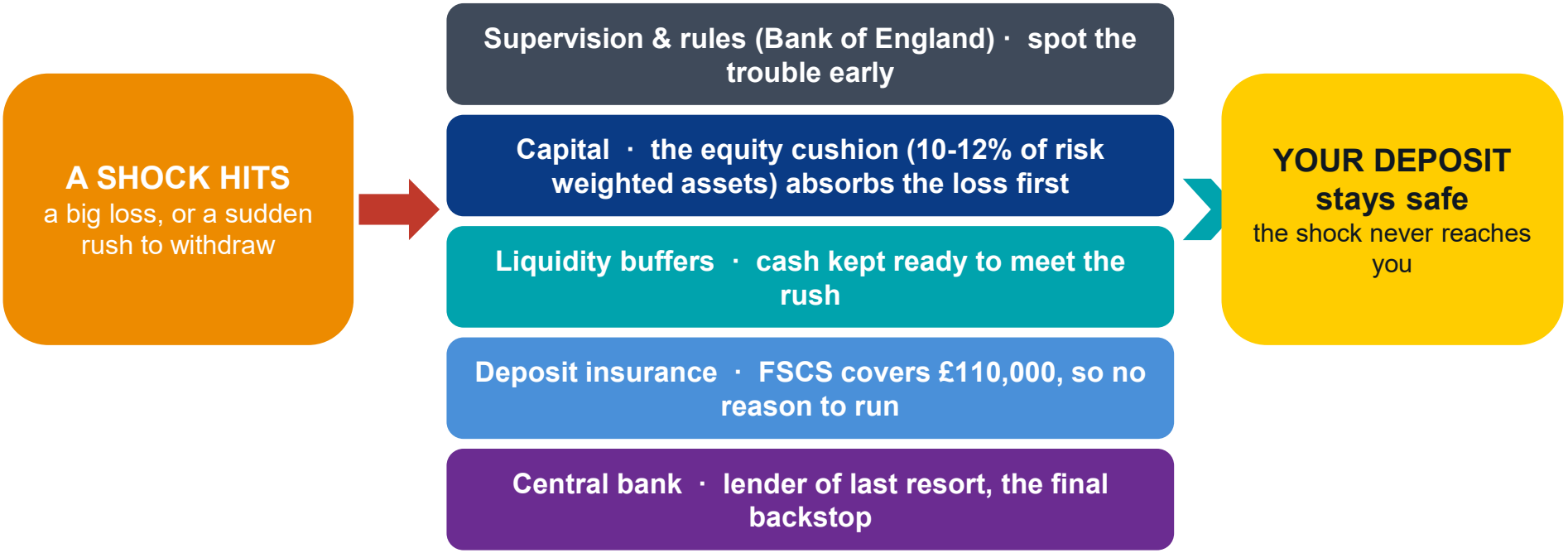
A balance-sheet problem. A loan just delays the end.

Bottom line: *"A classic problem in financial crises is to distinguish illiquidity from insolvency. It is critical and very hard to do."* **Tim Geithner, US Federal Reserve**

The safety net: how we stop the runs

Layer upon layer, built after each crisis

Most shocks are absorbed by the first layers; the last ones exist for the rare, severe run.



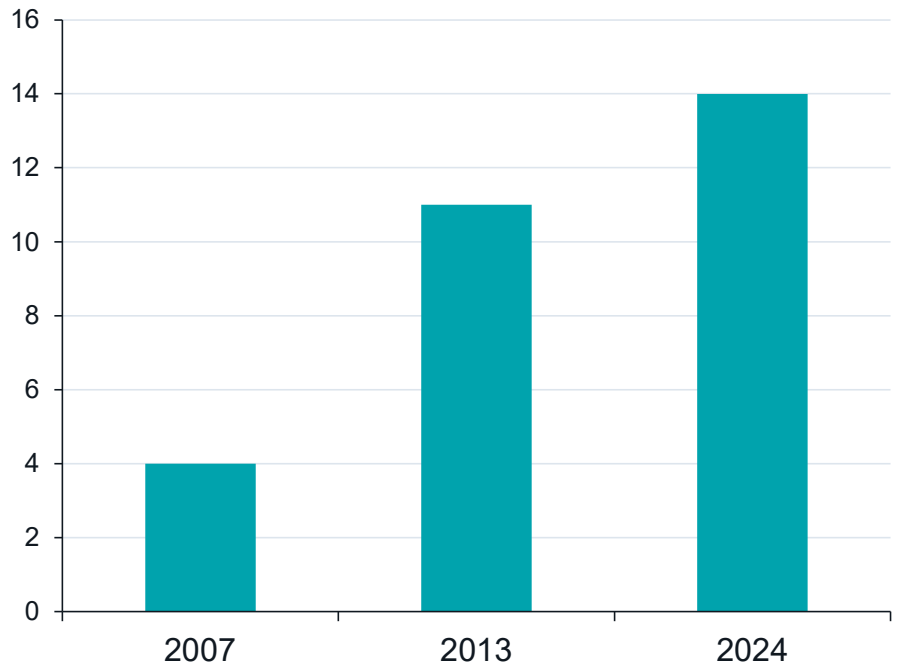
FSCS: Financial Services Compensation Scheme

Bottom line: deposit insurance stops the panic, capital absorbs the losses, the central bank is the backstop. Together they tame the fragility.

Have the fixes worked? Banks are far safer

Since 2008, the rules forced the equity cushion much thicker

Big-bank core equity capital % (approx)



The cushion roughly tripled since before the crisis.

Stronger, but not solved

After 2008, the Basel rules forced banks to hold far more equity, plus liquidity buffers and stress tests. The cushion roughly tripled.

So a bank today is genuinely much harder to topple than in 2007.

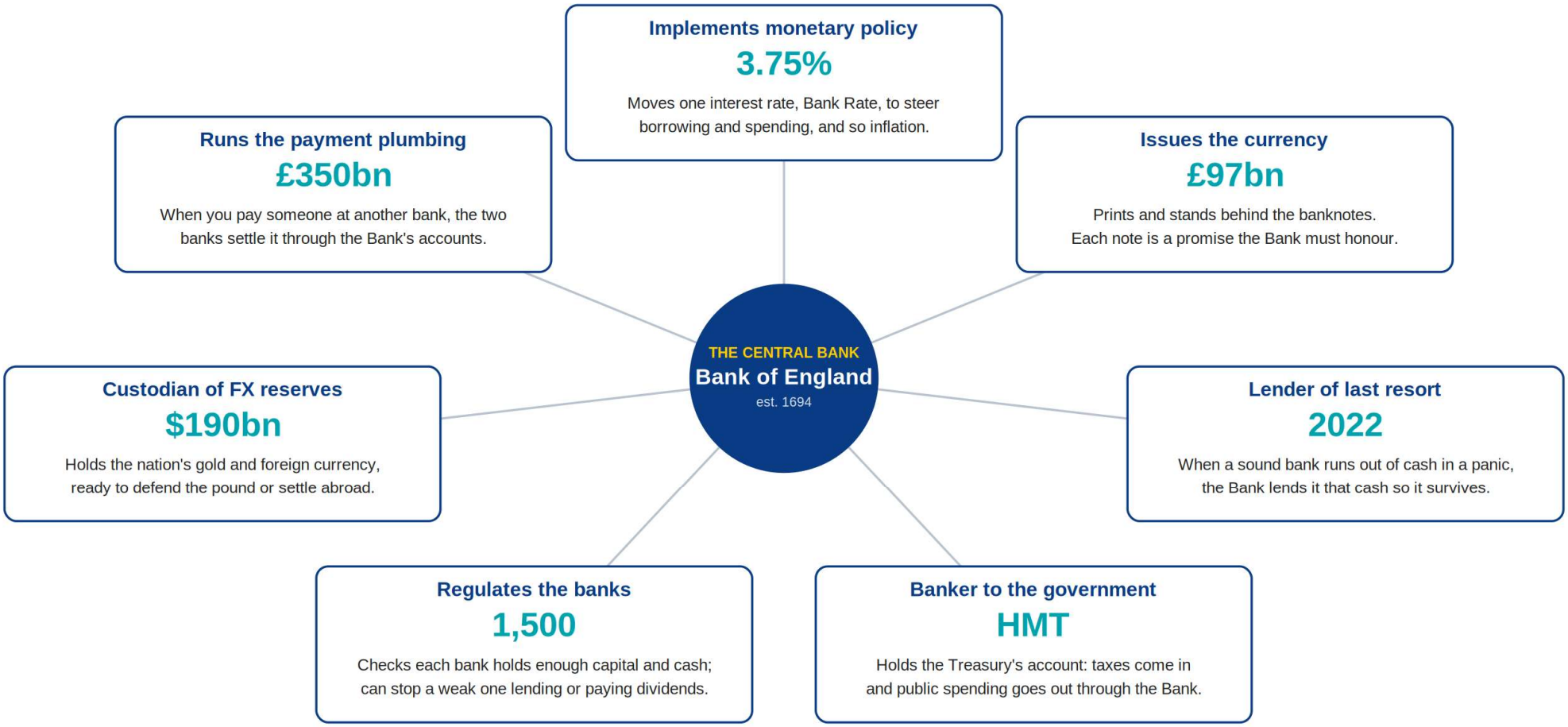
But 2023 proved it is not solved: mid-sized banks and the new SPEED of runs are the frontier, and the rules are still being fought over, Basel 3.1 tightening while others push to loosen.

Bottom line: Safer than 2007, but never 'safe'. The argument over how much capital is enough never ends.

Core capital (CET1 / equity), approximate, big banks: Bank of England, Basel Committee; 2007 figure is pre-Basel III tangible equity.

The central bank: seven jobs, one institution

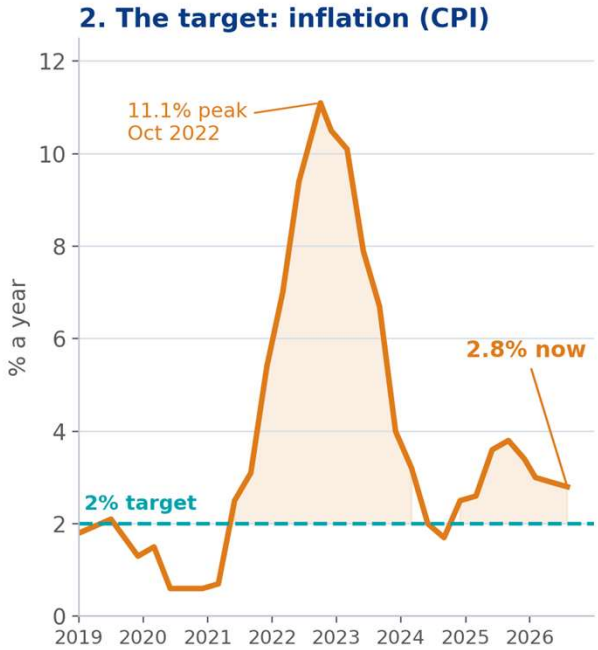
What the Bank of England actually does:

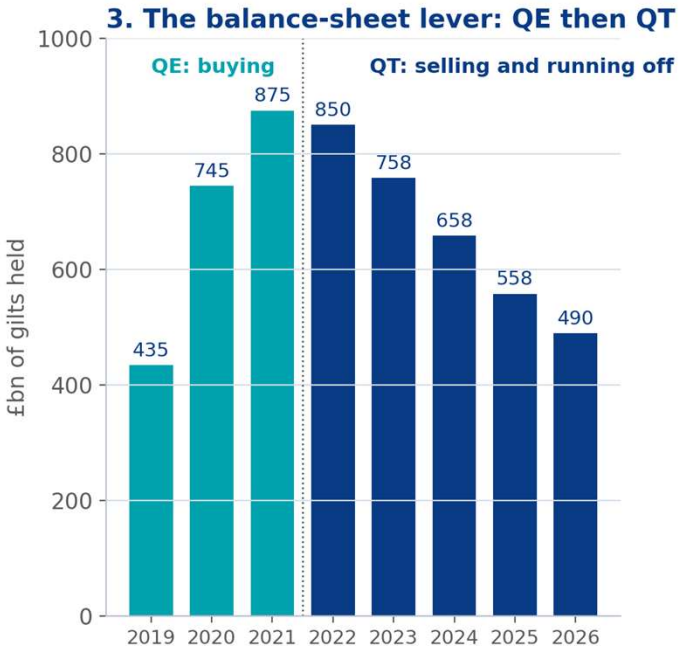


Monetary policy in action

Rates went from 0.1% to 5.25% to break 11% inflation, and are now coming back down







Sources: Bank of England (Bank Rate; Asset Purchase Facility gilt holdings, approx. end-Sept); ONS (CPI, selected months).

Quantitative easing and its reverse, quantitative tightening

The Bank creates new money and uses it to buy government bonds from pension funds and other big investors. Huge new demand pushes bond prices up, and since a bond pays a fixed amount, a higher price means a lower interest rate (£5 a year on £100 is 5%; on £125 it is 4%). Because mortgage and company borrowing rates are priced off the government bond rate, borrowing gets cheaper across the economy. It is used when Bank Rate is already near zero and cannot be cut further. **QT is the reverse:** the Bank sells the bonds back, or lets them mature, and the money comes back out of the system.

Three banks, one system

Same word, very different animals, and how the central bank sits behind them

	Commercial bank	Investment bank	Central bank
Takes your deposits?	Yes, its core business	No	No, only the banks' accounts
Creates money?	Yes, the ~97%	No	Yes: cash + reserves, the base
Its customers	You and businesses	Companies, governments, big investors	The banks, and the government
Inside the safety net?	Yes: insured, and the lender of last resort behind it	No, normally outside it	It IS the safety net
Regulated	Heavily; full banking licence	Lighter licence	It is the regulator

2008 proved it: Lehman, a pure investment bank, sat outside the net and was allowed to fail. Goldman Sachs and Morgan Stanley became bank holding companies overnight, to get inside the central bank's protection.

The other bank: the investment bank

No deposits, no high-street branches, it lives in the markets

How it differs

It takes no retail deposits, so no classic bank run, and a lighter licence.

Its clients are not you, they are companies, governments and big investors.

It does not lend from a deposit base; it connects issuers and investors in the capital markets.

Names: Goldman Sachs, JP Morgan, Morgan Stanley.

The two faces

Corporate-facing helping companies and governments RAISE money and do deals.

Investor-facing helping funds and big investors BUY, sell and trade those securities.

It stands between the two sides of the market, the classic sell side.

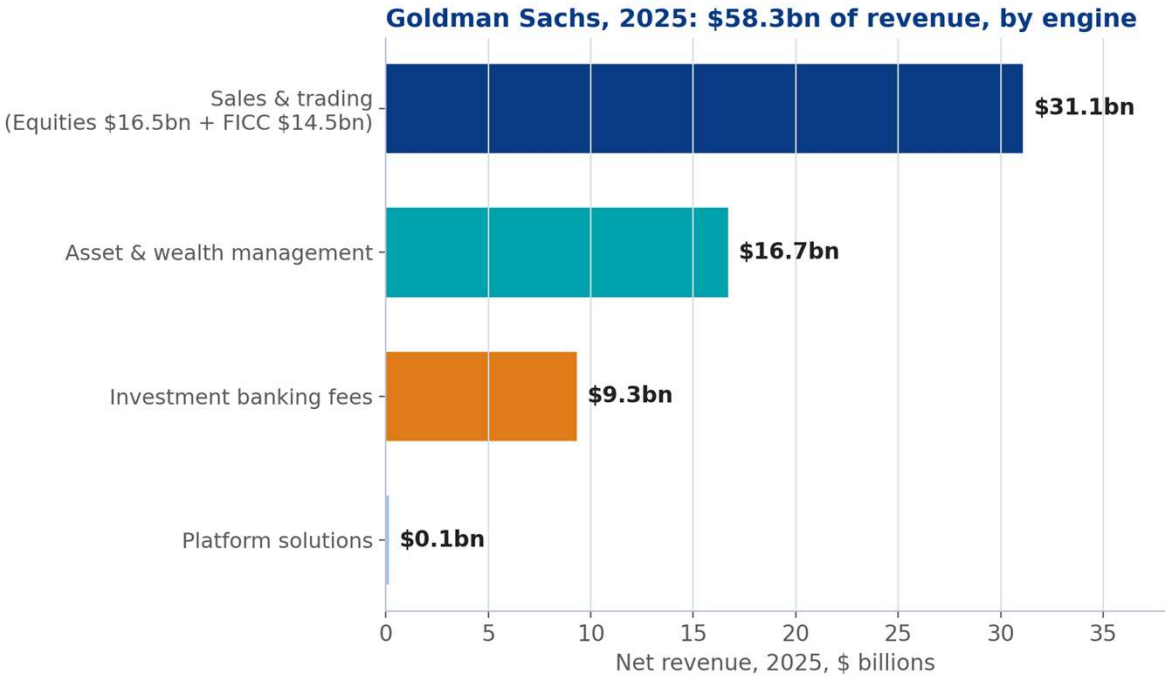
Bottom line: no deposits means no classic run, but its risks live in the markets, not on a high street.

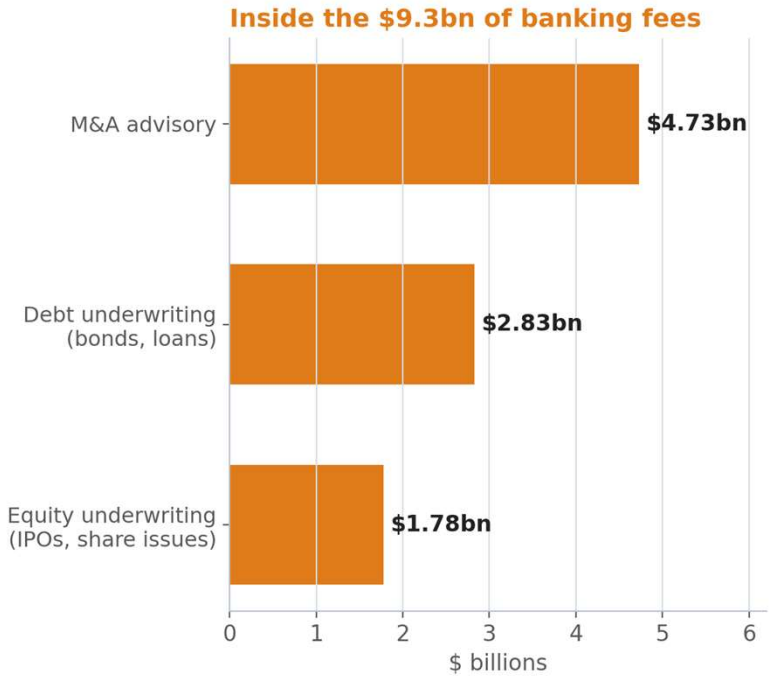
CLICK **What does an investment bank do? Ask AI** ↗

DEEPER READ **The global M&A boom rolls into 2026 as AI sparks a deal frenzy (CNBC, 2026)** ↗

Where the money really comes from

Goldman Sachs, 2025: the four engines, with real numbers





Sources: Goldman Sachs full-year 2025 results (SEC filing). Global investment-banking fee pool: $117bn in the first nine months of 2026, up 12% (LSEG).

The four engines: Raising capital = underwriting: take a company public or issue its bonds, and guarantee the sale for a fee. M&A advice = advising on buying and selling companies (this is my world). Sales & trading = buying and selling securities for clients (agency) and with the bank's own money (principal). Asset management = running clients' money for a steady fee.

The surprise: the glamorous deal fees are the smallest engine at $9.3bn; trading ($31bn) and managing money ($17bn) pay most of the bills. Globally the fee pool reached $117bn in the first nine months of 2026.

So you want to be an IB analyst?

A day on the desk, and six skills to start building this year

A day on the desk

first-year analyst, a live week

08:30 In early: overnight emails from New York and Asia; comps refreshed.

10:00 The model: filings and LSEG data in, valuation updated, every number tied out.

13:00 Pitch book: client slides, checked line by line by the associate.

15:30 Live deal: data room, buyer list, the client's due-diligence questions.

19:00 VP comments back: rebuild, re-check, re-send. Dinner at the desk.

23:00 Final turn to the MD. In a busy week, more comments after midnight.

The six skills it runs on, and how to build each one now

Top line: what the bank expects on day one. Teal line: what you can do this year.

Financial modelling

ON THE JOB

Build a three-statement model and a DCF from scratch, fast, with no errors.

BUILD IT NOW

Model one listed company end to end this term; check it against a broker note.

Accounting and valuation

ON THE JOB

Read a balance sheet in seconds; comps, DCF, the basics of an LBO.

BUILD IT NOW

This module and your accounting module, then the CFA Level 1 material.

The tools

ON THE JOB

Excel without a mouse, PowerPoint, LSEG Workspace, AI for drafts you then check.

BUILD IT NOW

Do the Workspace task every week; learn 20 Excel shortcuts; use AI, then verify it.

Attention to detail

ON THE JOB

Every figure ties out. One wrong number in a client book ends careers.

BUILD IT NOW

Check your own work twice before anyone else sees it, every single time.

Communication

ON THE JOB

A one-page summary a busy MD reads in a minute; speaking up clearly in meetings.

BUILD IT NOW

Write 200 words on one FT story each week; present in your seminars.

Commercial awareness, stamina

ON THE JOB

Know which deals are happening and why; long hours and hard deadlines.

BUILD IT NOW

Read the FT daily; get a spring week or internship; join the investment society.

Inside the bank: conflicts and walls

One roof, many clients, and information that must not leak

The conflict of interest

The same bank advises a company AND trades for investors AND bets its own capital.

Whose side is it on? It could misuse secret client information to trade for itself.

Principal trading (own capital) is powerful but risky, it helped sink banks in 2008.

The controls

'Chinese walls' strict information barriers between the deal-advising and trading sides.

Front / middle / back office the deal-makers, the risk controllers, and the settlement engine, kept separate.

Regulators watch for insider dealing and abuse.

Bottom line: one roof serving all sides only works while the walls hold. When they don't, you get scandals, fines and new rules.

The big picture: a wall built, torn down, and 2008

Why the line between the two kinds of bank keeps moving

1933	**Glass-Steagall** After the 1929 crash, the US legally SEPARATES commercial and investment banking. Deposits must not fund risky trading.
1999	**Repeal** Gramm-Leach-Bliley tears the wall down. 'Universal banks' now do both under one roof.
2008	**The reckoning** Risky investment-banking losses threaten deposit-taking giants. Taxpayers bail them out.
Now	**Ring-fencing** New rules again wall off everyday deposits from risky trading, to protect you and the system.

Bottom line: The eternal question: how far should everyday deposits fund risky markets? The argument is live right now, tighten (Basel 3.1) versus loosen.

CLICK **Should banks be split up again? Ask AI** ↗

The highlights, know these cold

1. A bank is licensed to create deposits, and deposits are ~97% of all money.
2. The magic trick: it transforms short, safe, small deposits into long, risky, large loans.
3. That makes it fragile: fixed liabilities, variable assets, and only ~5% equity.
4. Credit risk: expected loss = PD × LGD; expected losses hit revenue, unexpected losses hit equity.
5. A bank run is self-fulfilling; the key call is illiquidity (savable) vs insolvency (bust).
6. The safety net: capital (Basel), liquidity, deposit insurance (£120k), lender of last resort.
7. Three banks: commercial (does the trick), central (the backstop), investment (markets & deals).
8. Investment banks: raise capital, advise M&A, trade, manage assets; walls manage the conflicts.

Your mid-term multiple-choice test is built from this slide.

Exam check, can you answer these?

Same style as your mid-term test

A bank is 'illiquid' when it…

A) owes more than it owns B) can't find cash today but its loans are sound C) has no branches

Answer: B. Illiquid = a timing/cash-flow problem, not a value problem. That is insolvency.

A £1,000 loan has a 3% probability of default and 50% loss given default. Expected loss is…

A) £15 B) £30 C) £150

Answer: A. Expected loss = 3% × 50% × £1,000 = £15.

What makes a bank uniquely fragile?

A) it has too many staff B) fixed-value deposits on demand vs long, risky loans C) high taxes

Answer: B. Fixed liabilities on demand, funding long risky assets, on a thin equity cushion.

Which statement is TRUE?

A) an investment bank takes your deposits and lends them out B) the central bank is the 'lender of last resort'
C) commercial banks set Bank Rate

Answer: B. Investment banks take no deposits; the central bank is the backstop and sets Bank Rate.

Where we are, and where we're going

Lecture 3 of 5

1	**The big picture & the three pillars**	✓ done
2	**Balance sheets & financial intermediaries**	✓ done
3	**Commercial, investment & central banks**	✓ today
4	**Money & liquidity**	→ next week
5	Capital flows & the four sectors	

You now understand banks and the money they create. **Next: what money actually is, liquidity, and the new money, crypto and digital currencies.**

Over to you

Before next week

- Read Chapters 4, 7 & 8.
- Look ahead: Chapter 5 (money).
- Take the Blackboard quiz.
- Read one finance story in the FT this week.

Your Workspace task

1. Open a UK bank in LSEG Workspace (Lloyds, Barclays or NatWest).
2. Find total deposits and total equity on its balance sheet.
3. What % is funded by deposits? How thin is the equity cushion?

YOUR SEMINAR THIS WEEK **Seminar 3: Bank Structure & Credit Intermediation**

It goes deeper on today: the three types of bank, how they create credit, pricing risk (PD x LGD), and bank runs and regulation.

To prepare: read Chapters 4, 7 & 8.